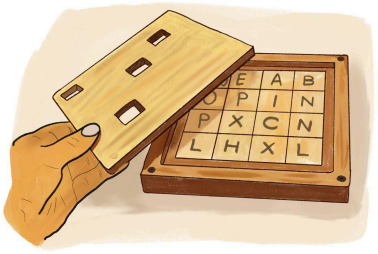
Rotating Grid Cipher

Grade 5: We the travellers II

Introduction

The need for secrecy when sending messages over the Internet is important for everyone. With this activity, students are introduced to the concept of basic cryptography/encryption and simple techniques to write the encrypted messages. Here, we will understand the Rotating Grid Cipher, use it to encrypt and decrypt a message. This activity is a very simple kind of encryption which was used during both the World Wars.



Activity	Time	Description
Launch	10 min	Start by asking students if they have ever shared a secret message with a friend. Ask questions like: How did you make sure others could not understand it? or Did you ever use code or a secret language? A teacher can even write an encrypted sentence on the board, and the students have to guess the message. Tell students that people also needed to send secret messages long ago. Introduce the idea that, even today, computers use special codes to hide information and keep messages safe on the Internet. Explain that in this activity, they will learn a simple way of hiding messages by turning a key pattern that has holes in it, called the Rotating Grid Cipher, and they will even make their own Rotating Grid Cipher to send secret messages to their friends. Demonstrate an example with the Rotating Grid Cipher. Template: https://tinyurl.com/rotatinggridtemplate Video: https://tinyurl.com/rotatinggridvideo

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Making and exploring the Rotating Grid Cipher by Students	15 min	Making and exploring the Rotating Grid Cipher Model and the Student's Tryout activity together, following the worksheet questions. Student Worksheet:
Discussions & Conclusion	10 min	Discussing the worksheets. Through this activity, students have learned how messages can be hidden using a Rotating Grid Cipher. We have discovered that encryption works by following a clear set of rules and that only someone who has the correct key (the key pattern and starting orientation) can decode the message.

CT Components

Algorithmic Thinking

We follow a step-by-step process to encrypt and decrypt messages: place the key pattern, write letters through the holes, rotate the key pattern over a grid, and repeat.

Decomposition

We can break down the encryption process into smaller steps, such as preparing the message, placing the key pattern, rotating it, and reading the final encrypted text. Also can break the larger problem (covering all 16 squares) into smaller parts by understanding how many squares one hole can cover.

Pattern Recognition

Observe how the holes move when the key patterns rotate, and how letters fill the grid systematically after each rotation. Also, observe the repeating pattern where every 90 degree rotation moves each hole to a new position, and after four rotations, the pattern repeats.

Generalisation

The same encryption rule can apply to different messages and grids of various sizes as long as the rotation pattern is followed. Larger rotations can be built from repeated smaller operations.

Abstraction

Focus on the rotation rule rather than the full grid details, and on how positions change systematically.

Evaluation

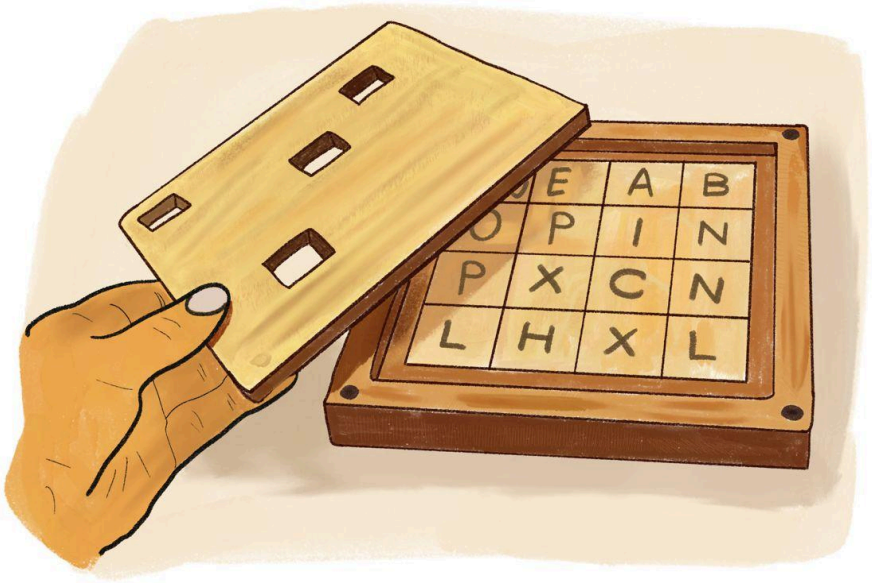
Whether their encrypted message can be correctly decoded by a friend, helping them verify if the key pattern and process were applied correctly. Also, test different key patterns and verify which one correctly reconstructs the original message.

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Activity: Turning Grid Cipher

People have used secret messages for centuries to protect important information. Even today, much of our communication online needs to be encrypted to keep our private data safe. For example, when your parents withdraw money from an ATM or make a UPI payment, they enter a PIN. This number needs to be hidden so that no one else can take their money, or when your parents buy something online and enter their bank card details, the information needs to be protected so that others cannot see or steal it.

A secret is information shared only with someone you completely trust through a message. Encryption is the process of changing a message so that its original meaning is hidden from anyone except the intended receiver. Though we use sophisticated methods to encrypt information today, we will understand encryption through a simple activity.



One simple way to do this is through ciphers. Cipher scrambles up the message using a **key**. To unscramble and access the message, you must have the key. This way, the cipher protects your message from those who do not have the key.

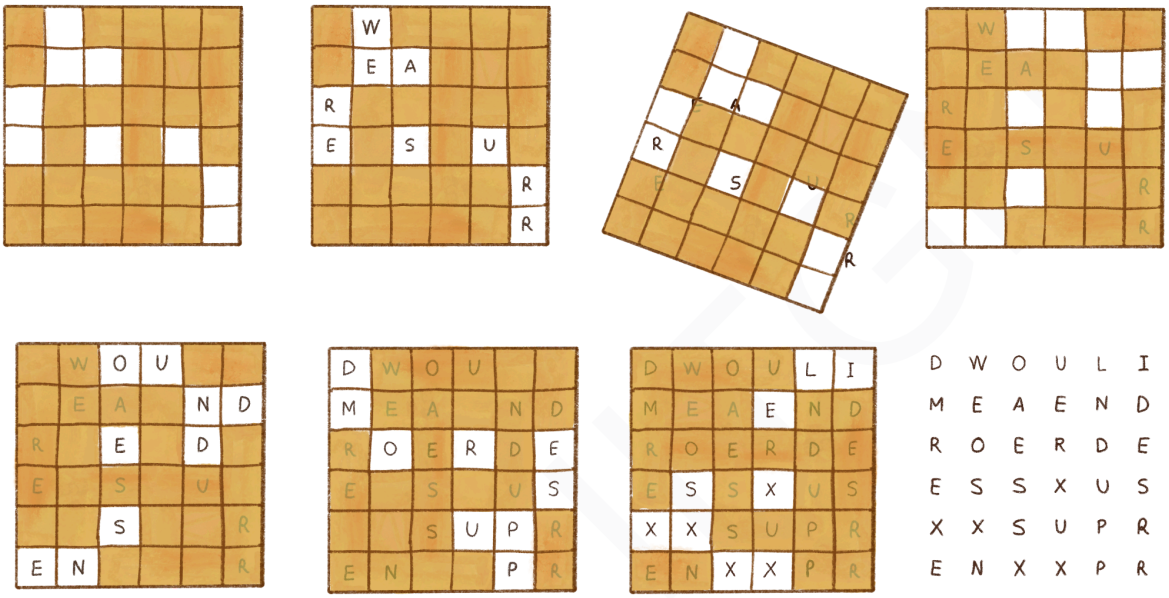
A well-known example of a cipher is the **Rotating Grid Cipher,** which was used during both World Wars. The Rotating Grid Cipher uses a square sheet with holes cut in it.

To encrypt a message, the sender must place the key pattern on an empty grid on paper and write letters of the message into the holes left to right, top to bottom. After filling all the holes, the key pattern is rotated clockwise by 90 degrees (the holes move to the empty spaces), and the process continues. After the key pattern is rotated 2 more times, there should be no empty spaces on the grid. This simple method allows messages to be hidden from anyone who does not know the key, the key pattern.

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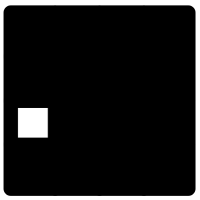
An example of the message "**WE ARE SURROUNDED. SEND MORE SUPPLIES** " is shown :

Original Message	WEARESURROUNDEDSENDMORESUPPLIESXXXXX
Encrypted Message	DWOULIMEAENDROERDEESSXUSXXSUPRENXXPR



In the above example, we have added 5 extra 'X' in the original message to make the total letters 36, matching the number of boxes in a 6 x 6 grid. We can also leave the box blank.

Q. A key pattern with a single hole for a 4x4 grid is shown here. How many different positions will a single hole visit as you rotate the key pattern 90-degree steps through a full turn?



a) 1
b) 2
c) 3
d) 4

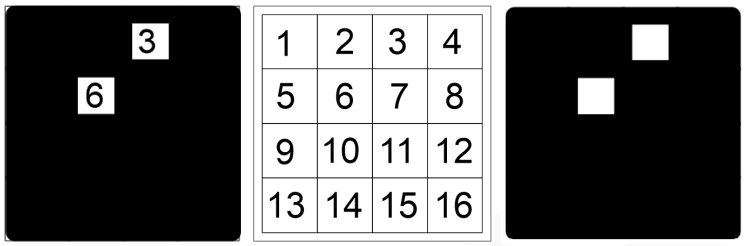
Explanation

d) 4

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When the key pattern is rotated by 90 degrees, the hole moves to a new position. If the key pattern is rotated by 90 degrees again, it moves to a new position. After four 90 degree rotations (90 degrees, 180 degrees, 270 degrees, and 360 degrees), the hole returns to its original position. Therefore, during one full rotation cycle, the hole will visit four different squares on the grid.
CT Competency: Pattern Recognition

Q. From a key pattern, the numbers 3 and 6 are visible. The key pattern is placed on a 4×4 grid numbered from 1 to 16. After rotating the key pattern 90 degrees clockwise once, which number will be visible through the hole of the key pattern?



a) 3 and 6
b) 7 and 12
c) 11 and 14
d) 5 and 10

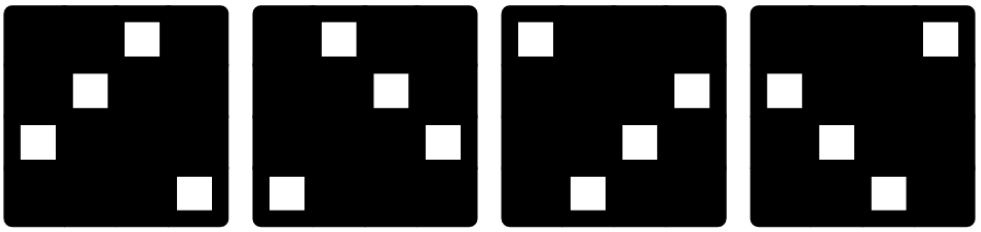
Explanation
b) 7 and 12
When the key pattern is rotated 90 degrees, each hole moves to a new location. Number 3 moves to number 12, and number 6 moves to number 7 after rotation.
CT Competency: Decomposition, Abstraction

Q. A 4×4 grid has 16 squares. When the key pattern is rotated four times (each time by 90 degrees), one hole shows 4 different squares. What is the minimum number of holes needed on the key pattern so that every square of the grid is visible once?

a) 1
b) 2
c) 4
d) 6

Explanation
c) 4
Each hole covers 4 different positions after the key pattern is rotated through all four orientations. Since the grid has 16 squares, dividing 16 by 4 tells us how many holes are needed. 16 ÷ 4 = 4, so four holes are required on the key pattern so that every square of the grid is filled exactly once during the rotations.
CT Competency: Decomposition, Abstraction, Generalisation

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When designing a key pattern, a basic rule for choosing hole positions is that the holes should cover every square on the grid using all rotations. Also, the initial orientation of the key pattern is important, as it gives the receiver a hint about which orientation to start the decryption process.

Q. Rotating a key pattern by 180 degrees is the same as rotating twice by how many degrees?

a) 0 Degrees
b) 270 Degrees
c) 180 Degrees
d) 90 Degrees

Explanation

c) 4

A rotation of 180 degrees means turning the key pattern halfway around. This can be achieved by performing two consecutive 90 degree rotations.

90 degrees + 90 degrees = 180 degrees. Thus, rotating twice by 90 degrees produces the same result as one 180 degree rotation.

CT Competency: Generalisation

Making of Rotating Grid Cipher

Print out the template file on A4 Paper for each student. Make pairs of students. (Guide Students to Make Rotating Grid Cipher)

Instructions for the activity

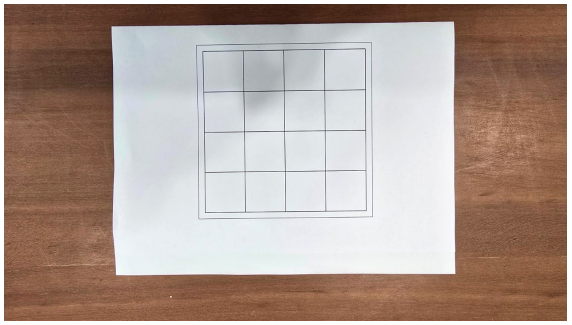
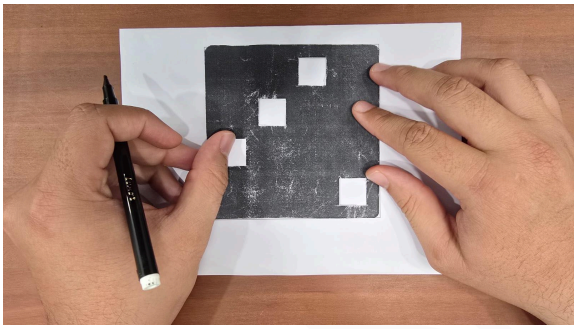
- We would cut out both squares from the template. One square is the 4×4 grid, which will be used to write the message. The other square is in black colour, and it will be used to create the key pattern.

- We will take the black square that has four smaller white squares, carefully cut out the white squares. We should now have a black square with four holes. This will be our key pattern.

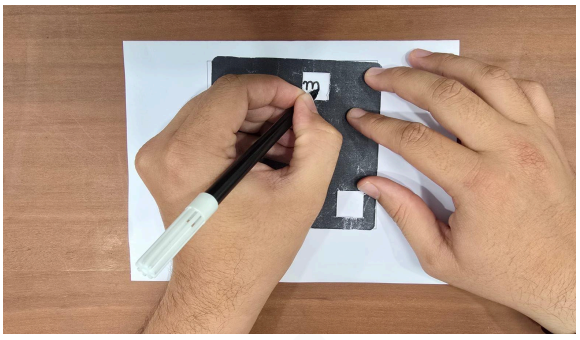
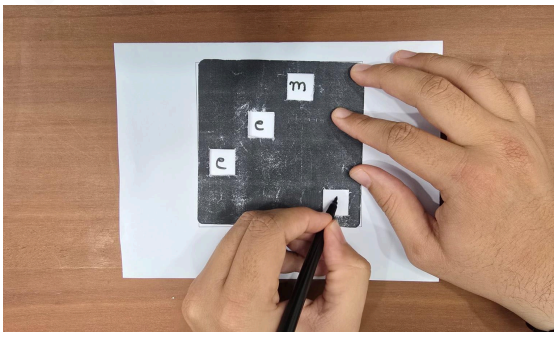
- We will write a short message with no spaces and up to 16 letters.
  For example: MEET ME AT TWELVE PM → MEETMEATTWELVEPM

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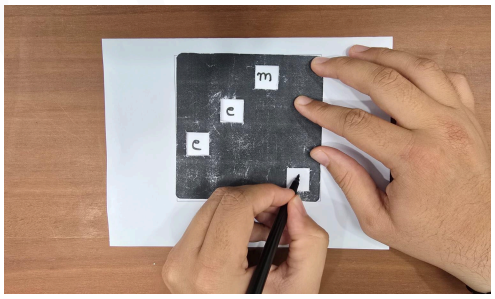
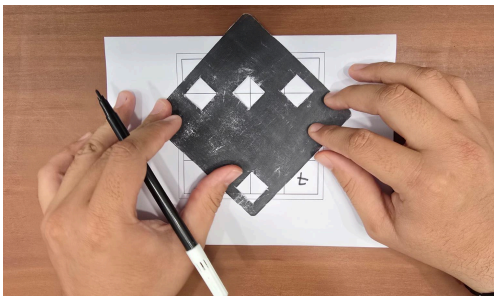
- We will place the key pattern over the blank 4×4 grid. We will remember the key pattern's initial orientation, because the same orientation will be needed to decode the message later.

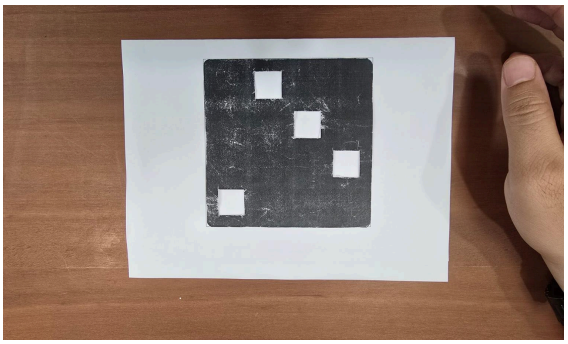
- Through the holes, we will write the first 4 letters of your message from left to right and top to bottom on the squares of the grid visible through the hole of the key pattern.
  Example: MEET

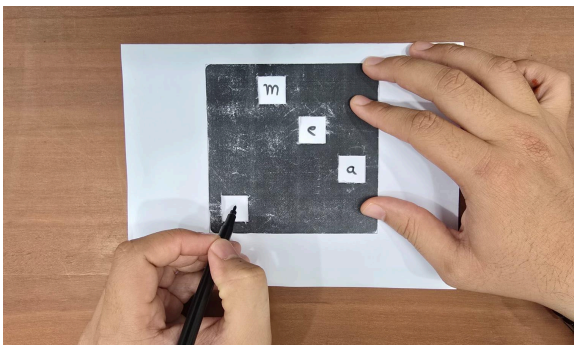



- Now, we will rotate the key pattern 90 degree clockwise and we will write the next 4 letters of the message on the squares of the grid visible through the hole of the key pattern.
  Example: MEAT

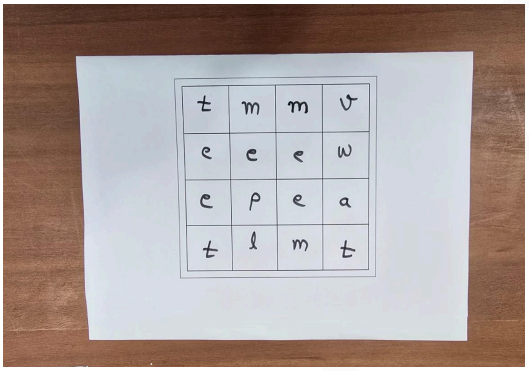



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- We will rotate the grid two more times, writing the next letters each time until all 16 letters are filled. The letters on the 4×4 grid will look random, When we remove the grid.



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- Now, we will convert the encrypted message on the 4x4 grid to a single-line message. We will write all the letters from the 1st row (left-to-right), followed by the second row, and so on. This will be our encrypted message.

- An example of the encrypted message is shown.

Message	MEETMEATTWELVEPM
Encrypted Message	TMMVEEEWEPEATLMT

- Once we have finished encoding our message, we will give the encrypted message and the Key Pattern (key) to our friend(s) to decode the message. From this, our friend(s) should first draw a 4x4 grid on a piece of paper and then write the received message on it, row by row (left to right, top to bottom).

- This process of decoding the message is similar to the method we used to encrypt the message.

- They will place the key pattern with its initial orientation over the message written on the grid.

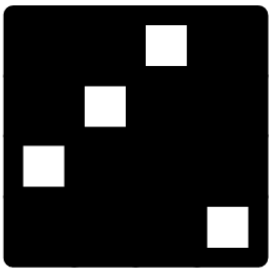
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- They will copy the letters visible through the holes (left to right, top to bottom) of the key pattern onto paper.

- They will now rotate the key pattern clockwise 90 degrees and copy the next 4 letters in the new visible holes next to the letters from the previous step.

- They will repeat the above steps for another two rotations to find all the remaining letters. This will be our original message sent to our friend(s).

Q. If the message "**APPLE IN LUNCH BOX**" is encoded over a 4x4 grid using the given key pattern. Which of the following is the correct encoded message?



a) UEABOPINPXCNLHXL
b) ABUEINOPCNPXXLLH
c) UEIBAPNNPOCHLXXL
d) UEBONAICPXHPNLXL

Explanation

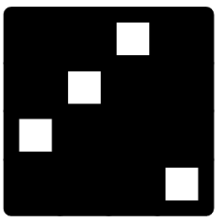
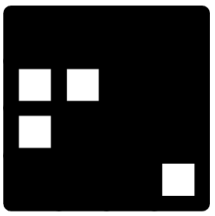
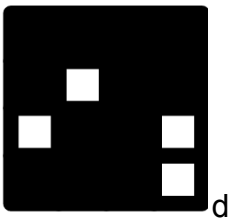
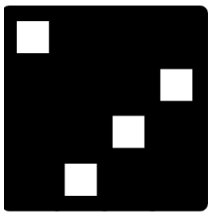
a) UEABOPINPXCNLHXL

First, we remove spaces from the message: APPLEINLUNCHBOX. This gives 15 letters. Since a 4×4 grid requires 16 letters, one extra X is added: APPLEINLUNCHBOXX

The key pattern is placed on the blank grid, and letters are written through the holes. After each set of letters is written, the key pattern is rotated 90 degrees clockwise, and the next set of letters is written. After four rotations, the grid becomes filled with letters. Copying the completed grid row-by-row produces the encrypted message: UEABOPINPXCNLHXL

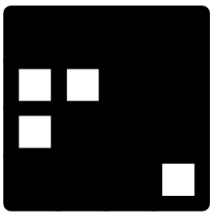
CT Competency: Algorithmic Thinking, Decomposition

Q. If the message "**APPLE IN LUNCH BOX**" is encoded over a 4x4 grid as "**UEIBAPNNPOCHLXXL**", which of the following grid patterns is used to encode the message?

a)  b)  c)  d) 

Explanation

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b) 

To determine the correct key pattern, we must reverse the encryption process.
We write the encrypted message into the grid row by row. Place each possible grid pattern on the grid. Rotate the grid in steps and read the letters through the holes. The correct pattern will reveal the original message APPLEINLUNCHBOXX when the grid is rotated.
CT Competency: Decomposition, Algorithmic Thinking, Evaluation

Through this activity, we have learned how messages can be hidden using a Rotating Grid Cipher. We have discovered that encryption works by following a clear set of rules and that only someone who has the correct key (the key pattern and starting orientation) can decode the message. This simple activity introduces the idea of cryptography and shows how computers protect sensitive information, such as passwords, ATM PINs, and online transactions, using more advanced encryption techniques.

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